

idk what im even learning here what the helly

chapter 1 Professional Ethics

ethical shield framework

1. Preventive design: predict & prevent ethical risks before it reaches end users
2. process safeguards: conduct ethical checks during dev cycle
3. accountability mechanisms: ensure transparency and answerability
4. cultural foundations: foster an environment where ethical behaviour is encouraged

ethical frameworks

1. consequentialism: how can i do the most good?
 - about end results
 - can ignore needs of smaller groups
2. deontology (duty): whats the right thing to do?
 - following the rules to the letter
3. virtue ethics: what kind of person should i be?
 - used when rules are not clear

principle of double effects (pros and cons)

If an action has good and bad outcomes, it is permissible to perform the action if the person intends for the good outcome and takes steps to avoid the bad outcome.

chapter 2 ai ethics and governance

ai ethics

1. privacy and surveillance
2. bias and discrimination
3. human judgement (in which case are humans required to step in)

4 types of ai

1. Reactive Machines

These AI systems act purely on the present situation. They cannot form memories or learn from past experiences, meaning they will always respond to identical situations in the same way.

2. Limited Memory AI

This type of AI can look into the past to inform its future decisions. It uses pre-programmed data or data collected from recent interactions to make more informed predictions and responses.

3. Theory of Mind AI

This is a conceptual type of AI that does not yet exist. It would be able to understand human thoughts, emotions, intentions, and beliefs, and be capable of forming opinions and social behaviors.

4. Self-Aware AI

The ultimate future stage of AI. This theoretical AI would be capable of having its own consciousness, emotions, and needs, and would possess a true understanding of itself and its place in the world.

model ai governance framework

1. Decisions by AI should be explainable, transparent and fair
2. AI systems should be human centric

ethical ai product management

BCI: brain-computer interface HCI: human-computer interface

Core ethical principles:

1. Impact on users
2. Trust and reputation
3. Legal implications
4. innovation and sustainability
5. professional growth
6. social responsibility

Product lifecycle:

1. Inception
2. Design and development
3. Verification and validation
4. deployment
5. operation and monitoring
6. re-evaluation
7. retirement

AI across industries:

- Edtech

- Healthcare
- FinTech
- Generative AI

Business model: economic, environmental and societal value

Dark patterns in products

Deceptive or manipulative design patterns to benefit company at expense of the users.

1. Fake Urgency: 'only n items left'
2. Sneak into basket: adding items to cart without consent
3. Hidden costs: unexpected costs added at the last step (eg. budget flights service fees)
4. Forced continuity: cannot cancel easily
5. Confirmshaming: uses guilt to manipulate users
6. Roach motel: easy to sign up, hard to get out of

3. UX/UI in HCI

1. user: needs, cognitive ability, emotions, diversity
2. interface: gui, haptic touch (vibrations, touch, sound)
3. task: usability (time taken, effectiveness)

Guidelines for design

1. put users first
2. design for everyone
3. be transparent
4. fight bias
5. earn trust

4. Automation and autonomy

automation: fixed set of rules, no adaptability (if else statements) autonomy: add AI, respond dynamically to environment

components of automation systems

1. sensors: receive input from environment
2. controllers: brain of machine
3. actuators: physical output
4. Human machine interface: control panel

components of autonomous system

1. perception
2. decision making
3. learning
4. action

excessive autonomy

- supervised automation: applications in war and self driving

eg. patriot missile shot down ally aircraft in fatal accident => needs supervision

- autonomy matrix: 2x2 matrix – low/high autonomy, low/high impact

autonomous healthcare

- dignity preservation: should not infantilize patients

- effect on the workforce: displacement of job
- anthropomorphism: robots mimicking human behaviour is confusing
- social justice: who gets access

governance

1. technical: validate/verify new models
2. ethics: resolve moral uncertainty
3. public acceptance: broader social implications (eg user's time)
4. professional responsibility: foster culture of safety
5. regulation: how to regulate products to be adaptive
6. oversight: address liability and oversee large deployments

data protection

collect minimum required data!!!

around the world

- sg: pdpa
- europe: gdpr
- california: ccpa

data lifecycle

1. collection
2. storage
3. usage
4. retention
5. deletion

2 ways to protect

1. anonymisation: redact sensitive stuff lmao
2. pseudonymisation: replace with pseudonym, can be reversed with a key

social engineering

1. phishing: by mail/sms/phone
2. smishing: by sms
3. vishing: by phone

data breach process

1. data incident suspected
2. investigation
3. notify pdpc (within 3 days)
4. notify users (as soon as practisable)

Intellectual property rights

1. copyright: no need to register with the govt, ip lasts for 70 years after death
2. patent: inventions, 20 year monopoly upon invention
3. trademark: signature/logo, 10 year upon registration

types of ip

- literary works: eg code
- dramatic works: eg plays, dramas

- artistic works: eg ui
- performances: eg broadcasts, show

challenges

1. monopoly & access
2. freedom of use (eg in learning)
3. cultural appropriation
4. ai and bias

how to register a patent

1. check registration criteria
2. apply for patent
3. preliminary examination
4. publication
5. search and examination
6. receive notice for eligibility of grant
7. file for issuance of certificate of grant
8. done

special pdpa

definitions

- individual: 1 person
- organisation: group of ≥ 1 person(s)
- personal data: data that can identify an individual
 - from the data
 - from the data and other info an organisation may have
- data intermediary: org which processes personal data on behalf of another org
- business contact info (BCI):
 - Name
 - Position or title
 - Business telephone number
 - Business address
 - Business electronic mail (email) address
 - Business fax number

PDPA covers data protection and do not call provisions.

data protection

1. collect, use and dispose
2. care of PD
3. individuals autonomy

obligations

- consent obligation (user must consent)
- purpose limitation obligation (only collect necessary data)
- notification obligation (notify user about how data is used)
- accuracy obligation (accurate info)
- protection obligation (secure data from unauthorized access)

- retention limitation obligation (retain data for as long as necessary)
- transfer limitation obligation (transfer out of singapore must comply with PDPA)
- access and correction obligation (allow users to request to correct their data)
- accountability obligation (appoint DPO)
- data breach notification obligation (alert user for cyber incidents)

special (ai ethics)

Definition

- Transparent
- Explainable and Interpretable
- Fair
- Auditable

do machines have minds

1. deductive: predicting the outcome (deterministic, if initial conditions are same, final result is same)
2. inductive: predicting the outcome (if initial conditions are same, final result is **probably** same)
3. abductive: best guess